Using Cascaded Jaynes-Cummings Systems to Distinguish Between Pulses With One and Two Photons.

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(Dated: November 2023)

The investigation of photon-photon interactions in quantum nonlinear optics is an important, highly studied field of physics, it has significant applications to quantum information processes. Nonlinear optics provides the opportunity to develop photon sorters, which promise to allow researches to perform deterministic Bell measurements. This paper investigates the photon sorting capabilities of two level atoms and atoms coupled to cavities following the Jaynes-Cummings model (JCM). In particular we use computational techniques to investigate the sorting capabilities for Gaussian and exponential pulses with different pulse widths. Our results confirm the photon sorting capabilities of a pair of atoms coupled to a unidirectional wave guide at low pulse widths $\sigma\gamma < 2$. We find that the results for Gaussian and exponential pulses differ quite significantly, demonstrating the sensitivity of this style of photon sorter to initial pulse shape.

I. INTRODUCTION

Nonlinear optics is the regime in which the superposition principle of electromagnetism does not hold, that is EM waves can interact. Nonlinear optics has found many applications in science and technology, including optical modulation and switching, nonlinear spectroscopy and frequency conversion [1]. The goal of quantum nonlinear optics is to develop systems in which photons can interact with each other. Significantly, quantum nonlinear optics allows for the development of quantum logic operators, and is used throughout quantum information processes. There have been a number of realisations of quantum nonlinear optics over the last decade (see for example [2]). One of the most important quantum nonlinear optical devices is the photon sorter [3–6]. Significantly, this device can be used to perform deterministic Bell state measurements which can be used in quantum computing, teleportation, repeaters, and cryptography [3, 4, 7]. [2–6] have demonstrated a variety of different methods to create photon sorters, including through the use of semiconductors, and two level systems.

We build on the work of [2] who has demonstrated the theoretical efficiency of using a pair of unidirectionally coupled two level systems (for example a two level atom) for photon sorting. We replicate the results of [2] using computational methods with different cost functions, and then extend our analysis to photon sorters that use a pair of atoms inside a cavity, following the Jaynes Cummings model (JCM). Specifically, we examine nanosecond Gaussian and Exponential pulses.

For Gaussian pulses, we find that a pair of atoms with small pulse widths ($\sigma\gamma < 2$) performs the best. We find that a single JCM has very poor results for CF1, and therefore cannot act as a photon sorter for Gaussian pulses. Additionally, we find that the two JCM sorter performs worse than the pair of atoms for small pulse widths, and that there is evidence that it may outperform the pair of atoms for high pulse widths. However, more investigation is needed. Contrary to Gaussian pulses, for exponential pulses we find the best performance is from a single JCM at low pulse widths.

Sec II establishes the necessary theoretical background for our analysis, including the Hamiltonians and Lindblad operators we use to model the system. Sec III explains our method, and the cost functions we used to asses efficiency of the photon sorter. Sec IV outlines our results. We finish with a discussion discussion of them in sec V.

II. THEORY

A. The Lindblad Master Equation

Quantum systems that interact with their environment can exhibit decoherence and dissipation. As a result, when dealing with these systems the use of deterministic equations is no longer feasible. Typically, the Lindblad Master Equation (Eq. 1) is used instead.

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_i \left(L_i \rho L_i^\dagger - \frac{1}{2}\left\{ L_i^\dagger L_i, \rho \right\} \right) \quad (1)$$

This equation describes the dynamics of the density matrix ρ, which represents a possible ensemble of different quantum states. Most significantly, this equation has the Lindblad terms L which represent how the system interacts with its environment. An accessible derivation of this equation for undergraduates can be found in [8].

B. Cascaded Systems

We are interested in systems in which the field moves from one subsystem to another but cannot move backwards. In practice this can be implemented using a wave

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guide. To develop a theoretical formulation, one considers two, separate, driven systems, where the second system is driven by the output from the first system. Doing so, we get the result that

$$L = \sqrt{\gamma_1}c_1 + \sqrt{\gamma_2}c_2$$

$$H = H_{sys} + \frac{1}{2i}\sqrt{\gamma_1\gamma_2}\left[c_2^\dagger c_1 - c_1^\dagger c_2\right]$$

Where $c_{1,2}$ are the first and second system operators. A detailed derivation can be found in [9]. This can be viewed as a coherent radiation term from L and a interaction term in H describing system one loosing energy while system two gains it, and visa versa. The unidirectionality arises from the interaction between these terms.

C. Input Traveling Pulse

We are interested in examining systems in which the input field is a traveling pulse of some shape $u(t)$. A technique for analysing such behaviour has been developed by [10], in which a cavity with some number of photons in it is cascaded into the system with some time varying coupling given by Eq. (2)

$$g_u(t) = \frac{u^*(t)}{\sqrt{1 - \int_0^t dt' \left|u(t')\right|^2}} \tag{2}$$

1. Exponential Pulses

For exponential wave pulses $u(t) = \sqrt{\lambda}e^{-\lambda t/2}$ we have that $g_u(t) = \sqrt{\lambda} = g_u$. So, we will proceed by writing exponential pulses as

$$u(t) = g_u e^{-g_u^2 t/2} \tag{3}$$

2. Gaussian Pulses

For Gaussian pulses

$$u(t) = \frac{1}{\sqrt{\sigma\sqrt{\pi}}}e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} \tag{4}$$

$g_u(t)$ cannot be found analytically (erf). Additionally, for small values of σ the formula for $g_u(t)$ has computational convergence issues.

D. Jaynes–Cummings Model

The two level Jaynes–Cummings Model is described in the rotating wave approximation by

$$H_{JC} = \hbar\omega_c c^\dagger c + \hbar\omega_a\frac{\sigma_z}{2} + \frac{\hbar\Omega}{2}\left(c\sigma^+ + c^\dagger\sigma^-\right) \tag{5}$$

Here the first term represents the cavity energy, the second term has the atomic inversion operator σ_z and represents the transition from the exited to ground state of the atom. The final term is the interaction between the cavity and the atom. Typically $\omega_{c,a}$ are of such a frequency that these terms can be approximately ignored.

E. Combined System

The system we are interested in, is a cavity that contains some number of photons, releasing energy into a wave guide, which is sequentially coupled to a pair of atoms (either isolated or inside a JCM). See figure 1 for details

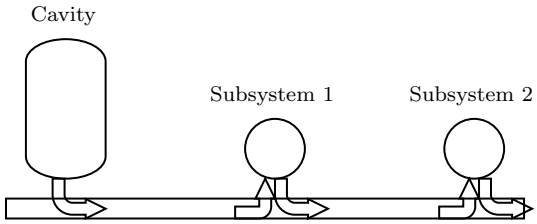


FIG. 1. Photon sorter setup. The subsystems can either be two level atoms, or JCMs.

Taking the initial cavity that emits the traveling pulse (Section II C) and cascading it into the first atom or JCM, and then further cascading this combined system into the second atom or JCM yields our final system of interest. It is also possible to introduce losses to the subsystems, and losses between the systems (see for example [2]).

Moving forward we will take $\hbar = 1$

1. Two Level Atoms

For the case of two atoms we have Eqns. (6) with c representing the input cavity with coupling $g_u(t)$ as described in Section II C, and $\sigma_{1,2}$ describing the atoms with coupling strengths $\gamma_{1,2}$.

2. Jaynes–Cummings Models

For the case of two Jaynes–Cummings Models we have Eqns. (7). Similarly, we have the input cavity c with coupling $g_u(t)$, the JCM cavities $c_{1,2}$ with coupling strengths $\kappa_{1,2}$, and JCM atoms $\sigma_{1,2}$ which couple to the $c_{1,2}$ via $\Omega_{1,2}$

$$L(t) = g_u^*(t)c + \sqrt{\gamma_1}\sigma_1 + \sqrt{\gamma_2}\sigma_2 \tag{6a}$$

$$H(t) = \frac{i}{2}\left(g_u(t)\sqrt{\gamma_1}c^\dagger\sigma_1^- + g_u(t)\sqrt{\gamma_1}c^\dagger\sigma_2^- + \sqrt{\gamma_1\gamma_2}\sigma_1^+\sigma_2^- - h.c.\right) \tag{6b}$$

$$L(t) = g_u^*(t)c + \sqrt{\kappa_1}c_1 + \sqrt{\kappa_2}c_2 \tag{7a}$$

$$H_{sys} = \frac{\Omega_1}{2}\left(c_1\sigma_1^+ + c_1^\dagger\sigma_1^-\right) + \frac{\Omega_2}{2}\left(c_2\sigma_2^+ + c_2^\dagger\sigma_2^-\right) \tag{7b}$$

$$H(t) = H_{sys} + \frac{i}{2}\left(g_u(t)\sqrt{\kappa_1}c^\dagger c_1 + g_u(t)\sqrt{\kappa_1}c^\dagger c_2 + \sqrt{\kappa_1\kappa_2}c_1^\dagger c_2 - h.c.\right) \tag{7c}$$

F. Output Traveling Pulse

Once the pulse has interacted with the system, we can use the quantum regression theorem to calculate the auto correlation function $g^{(1)}(t,t') = \langle L_s^\dagger(t)L_s(t')\rangle$ with $L_s(t)$ being the Heisenberg picture operator of L in Eqns (6,7). We have from [10] that $g^{(1)}(t,t') = \sum_i n_i v_i^*(t)v_i(t')$ where $v_i(t)$ are the output modes and n_i are the mean photon numbers. This factorisation can by calculated via the eigenvalues and eigenvectors of the matrix representation of $g^{(1)}(t,t')$.

III. METHOD

A. Cost Functions.

As outlined in Section II F, we can calculate the possible output modes $v_i(t)$ and their mean photon numbers n_i. There is only one possible output mode for the single photon, $v_{1p}(t)$. Let $v_{2p,i}(t)$ be the possible two photon modes. Ideally all the two photon modes would be orthogonal to the one photon mode, as such we define the cost function

$$CF_1 = \sum_i \tilde{n}_{2p,i}\langle v_{1p}|v_{2p,i}\rangle \tag{8}$$

Where $\tilde{n}_{2p,i}$ is the normalised mean photon number, which represents the probability of that mode being occupied. CF_1 can be interpreted as the expected overlap between the one and two photon outputs. We would also like the two photon output to be "single moded", hence we define

$$CF_2 = 1 - \tilde{n}_{2p,0} \tag{9}$$

B. Numerical Method

Equations 6 and 7 were implemented in QuTiP [11], and $g^{(1)}(t,t+\tau)$ was calculated. This was then converted to $g^{(1)}(t,t')$ using the fact that $g^{(1)}(t,t') = g^{(1)}(t',t)^*$. $v_{1,2p,i}$ and $n_{2p,i}$ were calculated according to section II F and the cost functions Eq (8, 9) were computed. The program was multithreaded for increased computational efficiency.

1. Exponential Pulses

A detailed description of the process taken for the exponential pulses is available in [12].

2. Gaussian Pulses

The parameters used are outlined in table I.

TABLE I. Parameters used for the Gaussian pulse. Chosen according to [13]

Parameter	Value (GHz)
$\gamma_1/2\pi, \gamma_2/2\pi$	0.3
$\kappa_1/2\pi, \kappa_2/2\pi$	20.1
$\Omega_1/2\pi, \Omega_2/2\pi$	9.23

$g_u(t)$ was calculated as in equation 2 via numerical integration. Due to issues with the precision of the numeric integrator, it could not be calculated for values of $\sigma\gamma < 1.5$. As a result, 12 simulations were run with evenly spaced values of $\sigma\gamma$ between 1.5 and 5.6. The simulations were run for a duration of 15σ as this gave the system sufficient time to settle, allowing for the complete output pulse to be calculated. 151 evenly spaced time steps were used.

IV. RESULTS

Examining the results, it is clear that computational resolution may be an issue for exponential pulses with

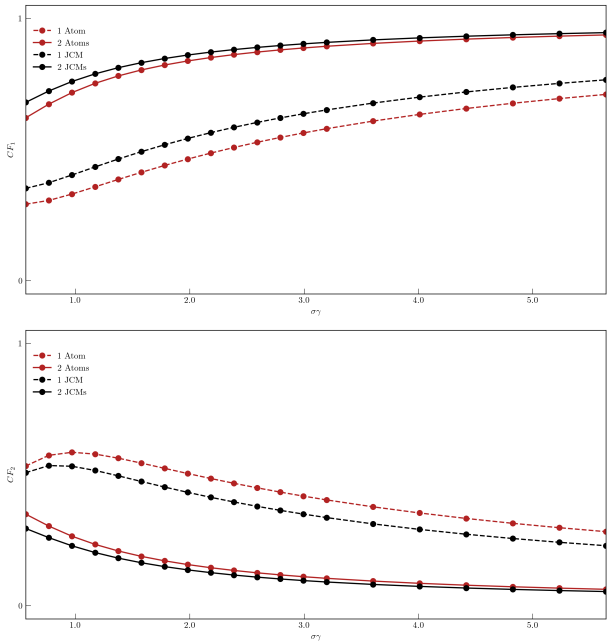


FIG. 2. **top**: The first cost function. **bottom**: The second cost function.
For exponential pulses, interacting with 1,2 JCMs, using parameters as in table I

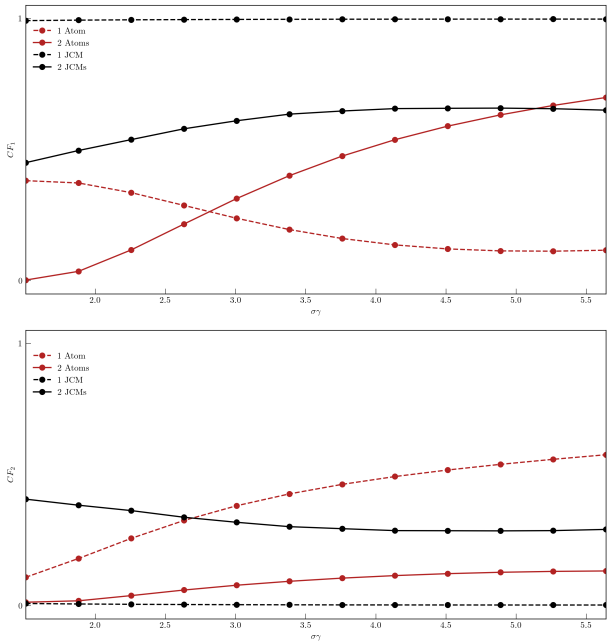


FIG. 3. **top**: The first cost function. **bottom**: The second cost function.
For Gaussian pulses, interacting with 1, 2 atoms and 1,2 JCMs, using parameters as in table I

low pulse widths. This could be fixed by increasing the number of time stamps from 151. However, the time complexity scales as $O(n^2)$ and increasing the number of time stamps significantly increases computational time (which is already multiple hours). While this may influence the precise values of the cost functions for low pulse widths exponentials, the "bulk" of the results are not significantly effected.

A. Exponential Pulses

The results for the Exponential pulses are in fig 2. CF1 is minimized by the single JCM at $\sigma\gamma = 0.6$, whereas CF2 is minimized by the pair of JCMs at $\sigma\gamma = 5.6$

B. Gaussian Pulses

The results for the Gaussian pulses are in fig 3. The first and second cost functions are minimized by the pair of atoms at a value of $\sigma\gamma = 1.5$. Contrary to the exponential pulse, the single JCM always performs poorly, and the pair of JCMs out performs the single JCM for CF1, but not for CF2, indicating a high dependence on pulse shape. This need further investigation.

V. DISCUSSION AND CONCLUSION

A note on experimental realisation. [2] discusses the experimental realisation of isolated two level atoms. The JCM can be realised according to [14]. It is likely that a single JCM would be setup, and the output pulse would be "fed" back into the same JCM after a sufficient amount of time to let the system settle. This would likely involve a high loss.

For Gaussian pulses, our results for the two atom case are in agreement with [2] providing merit to the procedure and the chosen cost functions. Our results have confirmed that a pair of two level atoms coupled to a unidirectional wave guide could behave as an efficient photon sorter for one and two photons. We find that the optimal value for the pulse width is $\sigma\gamma = 1.5$. Also in agreement with [2], we find that the two atom photon sorter looses efficiency with wider Gaussian pulses. Our results provide evidence that a pair of JCMs could potentially provide promising results as a photon sorter for sufficiently high pulse widths. However, further testing needs to be done to confirm this hypothesis. Our results also indicate that a single JCM makes a poor photon sorter for Gaussian pulses. We find that the results for the exponential pulse differs from that of the Gaussian pulse, demonstrating the systems sensitivity to pulse shape. However, we find that the results for the exponential and Gaussian pulses appear to follow a similar "shapes".

While it is possible to create the experimental subnanosecond pulses required for the two atom case (see for

example [15–17]), we have not examined the dynamics of real pulses produced from experiments where a photon sorter would be used. [2] Has examined the two atom system with loss and imperfections in the setup. The same needs to be done for the JCM setup.

Consequently, future work can go in 6 directions. (1) computing the result for Gaussian's with high pulse widths (2) computing low pulse width Gaussian's with the use of high precision numeric integrator (3) confirming the results for low width exponentials by using more time steps (4) testing different pulse shapes (5) improving the model with losses and real input pulses (6) experimental implementation of the setup.

VI. ACKNOWLEDGMENTS

I would like to thank my supervisor, Sahand, for giving up his time each week and providing guidence and insight for the topic. I would also like to thank Yuktee and Haoyuan for joining our meetings. Final I want to acknowledge the work Akito did for the exponential pulses.

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